

Technology and Profitability Evidence

Model performance, customer ROI and limits

Evidence	Observed result	Limit
Underwriting AUROC	0.84 vs 0.72 baseline	portfolio mix differs in two segments
Fraud precision at 80% recall	0.58 vs 0.41	validated on three customers
Pilot 1 annualized ROI	352%	signed customer confirmation
Pilot 2 annualized ROI	233%	six-month extrapolation
Pilot 3 annualized ROI	261%	startup methodology only
Equal opportunity gap	0.061 vs 0.048 baseline	startup model is worse, though below internal 0.08 threshold

Profitability conclusion

The technology has evidence of economic value beyond benchmark accuracy. Profitability remains contingent on customer conversion, retention, segment-level validation, and completion of compliance work without extending the sales cycle.